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Lecture 5: Transforms, Duality, Entropic OT

Reading

Peyré & Cuturi, *Computational Optimal Transport* [1]. We follow their notation and conventions unless noted.

1 Last Time

What we built. With $c : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$ and marginals α, β, the (continuous) Kantorovich primal is

$$\min_{\pi \in \Pi(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\, d\pi(x,y),$$

where $\Pi(\alpha,\beta)$ is the set of couplings with marginals α and β.

For $c(x,y) = d(x,y)^p$, the Wasserstein distance is

$$W_p(\alpha,\beta) \triangleq \left(\inf_{\pi\in\Pi(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} d(x,y)^p\, d\pi(x,y)\right)^{1/p}.$$

This metrizes weak convergence in many settings and endows $\mathcal{P}(\mathcal{X})$ with geometry [1, Chs. 2–3].

Dual viewpoint (informal). Feasible potentials f, g must satisfy $f(x) + g(y) \leq c(x,y)$; the (proper) dual is a supremum over such potentials:

$$\sup_{f,g:\, f\oplus g\leq c} \left(\int_{\mathcal{X}} f\, d\alpha + \int_{\mathcal{Y}} g\, d\beta\right).$$

This "price" view explains structure and yields certificates of optimality.

Q&A

- **Q:** The slides showed an "inf" in the dual. **A:** This was a typo in the slides. The dual maximizes a lower bound, hence sup.

2 c-transforms, c-conjugates, c-convexity

2.1 Motivation

The constraint $f(x) + g(y) \leq c(x,y)$ ties f and g together in a cost-aware way. Rather than optimize over two unrelated functions, we can fold one into the other via a *c-transform* that depends on the geometry encoded by c [1, §5.1].
Another way of thinking about it is that we have linear programming conditions within the OT dual problem. We also get a generalized notion of convexity that is determined by the cost $c(x,y)$ and the geometry it encodes. OT duality forces us to invent a cost-dependent geometry to express dual potentials.

2.2 c-transforms

Definition 1 (c-transform)**.** *Given $\phi : \mathcal{X} \to \mathbb{R}$ and cost c, define*

$$\forall y : \phi^c(y) \triangleq \inf_{x \in \mathcal{X}} \big(c(x,y) - \phi(x)\big).$$

Then $\phi(x) + \phi^c(y) \le c(x,y)$ for all (x,y), and ϕ^c is the "tightest" partner of ϕ without breaking feasibility.

- We break feasibility the moment that $f(x) > c(x,y) - g(y)$
- We want to maximize $f(x)$, and the largest we can make $f(x)$ before breaking feasibility is when $f(x) = \inf_y c(x,y) - g(y)$
- Analogously, best choice for $g(y)$ is $g(y) = \inf_x c(x,y) - f(x)$

Q&A

- **Q:** What does the small triangle over "=" mean in $a \triangleq b$? **A:** "Defined as."
- **Q:** In $\inf_x$, what is the subscript for? **A:** It indicates the variable(s) over which the infimum is taken.
- **Q:** The slides "flipped the arguments" and did not have a minus in front of ϕ ($\inf_{x \in \mathcal{X}} \big(\phi(x) + c(x,y)\big)$). **A:** Yes the standard convention is $\phi^c(y) = \inf_x \{c(x,y) - \phi(x)\}$ [1, §5.1], which is what we decided to include in our scribe notes.

2.3 c-conjugacy

Definition 2 (c-conjugate pair and c-convexity)**.** *A pair (f,g) is c-*conjugate *if*

$$\forall x : f(x) = g^c(x) = \inf_y \big(c(x,y) - g(y)\big), \qquad \forall y : g(y) = f^c(y) = \inf_x \big(c(x,y) - f(x)\big)$$

*A function f is c-*convex *if (f, f^c) is c-conjugate and same for (g, g^c); equivalently $(f^c)^c = f$, $(g^c)^c = g$ [1, §5.1].*

Reducing to one potential. The dual can be written as a supremum over c-convex functions:

$$\sup_{f \in c\text{-cvx}} \left(\int f \, d\alpha + \int f^c \, d\beta \right)$$

so we track one function f plus its transform f^c.

Two helpful special cases.

- $c(x,y) = \|x - y\|$: c-convexity $\iff$ 1-Lipschitz; $f^c = -f$ [1, §6.1].
- $c(x,y) = \|x - y\|^2$: c-convexity reduces to classical convexity, and f^c is the Legendre–Fenchel conjugate $f^*(y) = \sup_x \langle x, y \rangle - f(x)$ [1, §5.1].

Q&A

- **Q:** Does this reformulation constrain f and g too much? **A:** No; at optimum one can choose $g = f^c$ (or $f = g^c$), so restricting to c-convex f is without loss [1, §5.1].
- **Q:** Is c-conjugacy always possible? **A:** For the Kantorovich dual under mild assumptions on c and the spaces, yes. You can replace any feasible pair by a c-conjugate pair attaining the same value.

With this machinery, we only need to track one dual variable instead of two, but we need to find an efficient way to compute the conjugate. These conjugates appear very often in optimization.

3 Duality

3.1 Deriving the discrete dual (step-by-step)

Let $a \in \mathbb{R}_+^n$, $b \in \mathbb{R}_+^m$ with $\mathbf{1}^\top a = \mathbf{1}^\top b = 1$, and $C \in \mathbb{R}^{n \times m}$. The primal is

$$\min_{\pi \in \mathbb{R}_+^{n \times m}} \langle C, \pi \rangle \quad \text{s.t.} \quad \pi \mathbf{1} = a, \;\; \pi^\top \mathbf{1} = b.$$

Form the Lagrangian lower bound

$$H(f, g) \triangleq \min_{\pi \geq 0} \Big(\langle C, \pi \rangle + f^\top (a - \pi \mathbf{1}) + g^\top (b - \pi^\top \mathbf{1}) \Big).$$

Rearranging we get:

$$H(f, g) = f^\top a + g^\top b + \min_{\pi \geq 0} \langle \pi, \; C - (f \oplus g) \rangle, \qquad (f \oplus g)_{ij} \triangleq f_i + g_j.$$

The inner minimum equals 0 if $C - (f \oplus g) \geq 0$ (take $\pi = 0$), and $-\infty$ otherwise (send some $\pi_{ij} \to \infty$). Therefore,

$$\max_{f, g} H(f, g) = \max_{f, g} \{ f^\top a + g^\top b : \; f \oplus g \leq C \},$$

and strong duality gives

$$\min_{\pi \in \Pi(a, b)} \langle C, \pi \rangle \;\; = \;\; \max_{f \oplus g \leq C} (a^\top f + b^\top g).$$

Q&A

- **Q:** If $\pi_\star$ is optimal, why only $H(f, g) \leq \langle C, \pi_\star \rangle$? **A:** H is a lower bound obtained by relaxing constraints; equality holds at optimal duals.
- **Q:** Why does a negative entry in $C - (f \oplus g)$ force $-\infty$? **A:** The minimization over $\pi \geq 0$ is linear; a single negative coefficient can be exploited by sending the corresponding $\pi_{ij} \to \infty$.
- **Q:** Dual *sup* vs. *inf*? **A:** Dual is a *supremum* because it optimizes the best lower bound.

3.2 Implications

- Key takeaway is that duality is not just a computational trick, but a structural lens through which we can understand OT.
- The implication is that dual ≤ primal, but can have stronger duality: Dual ≤ Primal
- Guarantees no duality gap because dual solutions faithfully represent the transport cost
- Provides alternative characterization of OT distance (as sup over potentials)

4 Entropy Regularization

4.1 Motivations for Regularization

- Computationally expensive. For example, discrete OT is a linear program with $O(n^2)$ variables. Worst-case complexity is cubic, which is intractable/inapplicable for large n.
- However, even if we could solve it, we may not necessary want the solution. The optimal plan is often sparse and unstable. By regularizing, we are smoothing the plan out.
- The idea is to add an entropy term to smooth solutions and encourage more diffuse transport plans. It also makes optimization strictly convex.

4.2 Entropy of π and Formulation

Define $H(\pi) = -\sum_{i,j} \pi_{ij} \log \pi_{ij}$, which will be our regularization term. The entropic OT (EOT) problem is

$$\min_{\pi \in \Pi(a,b)} \langle C, \pi \rangle - \varepsilon H(\pi), \qquad \varepsilon > 0.$$

Heuristics. Small $\varepsilon \Rightarrow$ closer to exact OT but rougher; large $\varepsilon \Rightarrow$ blurrier plan, faster and stabler.
The intuition here is that high $H(\pi)$ means that there are many mass-to-mass transfer happening, which means more mass splitting.
Unique Observations.

- The term $\epsilon > 0$ represents the regularization strength.
- Alternatively, we can use relative entropy, e.g. using KL divergence.
- Because it is strongly convex, there is a unique solution.
- Differentiable with respect to inputs (α, β, C)

Q&A

- **Q:** Can we learn with large ε then refine? **A:** Yes, we can first learn how to solve the entropy problem fast and use those results as a warm-start for fine-tuning for the actual solution.

Let $K_{ij} = \exp(-C_{ij}/\varepsilon)$. The unique optimizer has the scaling form

$$\pi_\varepsilon \;=\; \mathrm{diag}(u)\, K\, \mathrm{diag}(v), \quad \text{with} \quad \pi_\varepsilon \mathbf{1} = a, \;\; \pi_\varepsilon^\top \mathbf{1} = b,$$

for positive vectors u, v determined by the marginal constraints [1, §4.2]. As $\varepsilon \to 0$, π_ε approaches an OT optimizer (selecting the max-entropy one when multiple exist); as $\varepsilon \to \infty$, $\pi_\varepsilon \to a \otimes b$ [1, §4.3].

Q&A

- **Q:** Are there any magic numbers for ε? **A:** There's no universal choice since it depends on the nature of the data. We can tune by validation; for example, denser/noisier data tolerate larger ε.
- **Q:** Since a full sweep over ϵ is expensive, is there any way to bypass that? **A:** Once again it depends on context of problem. You can do binary search for faster computation. Additionally, if your algorithm fails because of numerical instability, it is a good sign that you are regularizing too much.
- **Q:** Is information in K alone sufficient? **A:** Not for the final plan: u, v must balance to enforce marginals; K is only the Gibbsized cost.

4.3 Sinkhorn's Algorithm

- **Fixed-point equations.**

$$a \;=\; \pi_\varepsilon \mathbf{1}_m \;=\; u \odot (Kv), \qquad b \;=\; \pi_\varepsilon^\top \mathbf{1}_n \;=\; v \odot (K^\top u),$$

  where $\odot$ denotes elementwise product.

- **Iterative scaling (Sinkhorn).**

Algorithm 1: Sinkhorn (matrix scaling) for EOT

Input: $C \in \mathbb{R}^{n \times m}$, $a \in \mathbb{R}^n_+$, $b \in \mathbb{R}^m_+$, $\varepsilon > 0$

$K \leftarrow \exp(-C/\varepsilon)$

Initialize $u \leftarrow \mathbf{1}_n$, $v \leftarrow \mathbf{1}_m$

while *not converged* **do**

  $u \leftarrow a \oslash (Kv)$

  $v \leftarrow b \oslash (K^\top u)$

return $\pi_\varepsilon \leftarrow \mathrm{diag}(u)\, K\, \mathrm{diag}(v)$

Each iteration is $O(n^2)$; typically tens/hundreds of iterations. Log-domain tricks improve stability [1, §4.4]. Note: $OT_\varepsilon(a, a) \neq 0$ (bias); "Sinkhorn divergences" debias this while preserving efficiency (see 1, §5 for discussion).

4.4 Richard D. Sinkhorn

Sinkhorn (1964) analyzed convergence of alternating row/column scaling to achieve prescribed marginals, now foundational for EOT solvers. The method also appears as IPFP/RAS in statistics/economics (historical notes in 1, Ch. 4).

4.5 Sinkhorn: Projection View

EOT can be seen as projecting the Gibbs kernel K onto the transport polytope using the Kullback–Leibler divergence:

$$\pi_\varepsilon \;=\; \arg\min_{\pi \in \Pi(a,b)} \mathrm{KL}(\pi \,\|\, K).$$

Sinkhorn performs alternating KL-projections onto the affine sets $\{\pi : \pi\mathbf{1} = a\}$ and $\{\pi : \pi^\top \mathbf{1} = b\}$, which explains its monotone progress and robustness [1, §4.4].

4.6 Sinkhorn: Historical Perspective

From early contingency-table fitting to modern large-scale ML, the same scaling idea reappears with different names. Entropic OT ties these threads together and provides a principled objective whose optimizer is exactly what scaling computes [1, Ch. 4].

4.7 Implications of EOT

- **Computational**: Dramatically improves scalability. GPU-friendly. Handles large datasets
- **Optimization-friendly**: Entropic regularization makes OT loss smooth and differentiable. Gradients propagate through Sinkhorn iterations. Enables end-to-end learning with OT layers
- **Statistical**: Unbiased correction of entropic OT cost. Balances comp. efficiency with better statistical convergence

4.8 Applications in ML

- **Generative modeling.** Sinkhorn GAN: uses entropic OT as a training loss, leads to faster, stable optimization. Provides a differentiable surrogate to Wasserstein distance.
- **Representation learning.** Align embeddings across languages or modalities with regularized OT. Benefits include couplings spread mass to avoid overfitting to noise.

References

[1] Gabriel Peyré and Marco Cuturi. Computational Optimal Transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019. (See also the arXiv preprint arXiv:1803.00567.)